

Problem 1:

a) You are given a scalar field f and a vector field $\mathbf{A}$, such that $\mathbf{A} = \nabla f$. What is $\nabla \times \mathbf{A}$?

For each of the two vector fields in b) and c) below, calculate the curl and divergence. Then circle the correct answer to whether the field is conservative, solenoidal, both, or neither.

b) $\mathbf{A} = \hat{\mathbf{x}}(2x^2 + 5z^2) - \hat{\mathbf{y}}(2y^2 + 5x^2) + \hat{\mathbf{z}}(2z^2 + 5y^2)$

conservative

solenoidal

both

neither

c) $\mathbf{A} = \hat{\mathbf{R}}5\exp(-10R)/R$

conservative

solenoidal

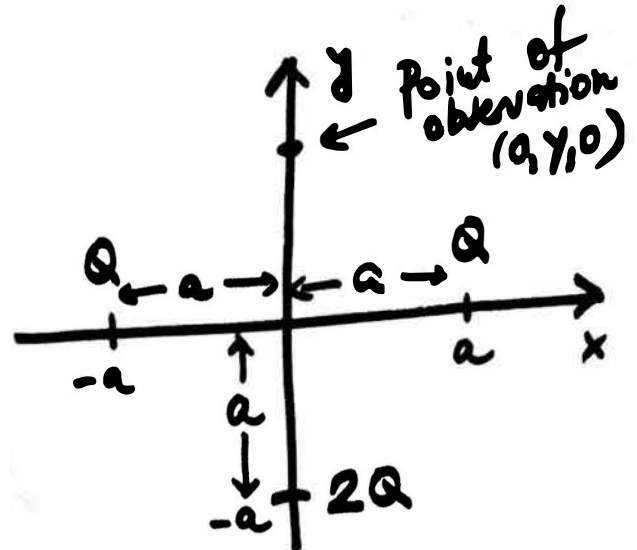
both

neither

Problem 2:

a) Calculate the electric field $\mathbf{E}$ along the y axis due to the distribution of point charges depicted on the right.

Assume the surrounding medium is air (permittivity ϵ_0).



b) Calculate the electric potential V along the y axis. Assume $V(0)=0$.

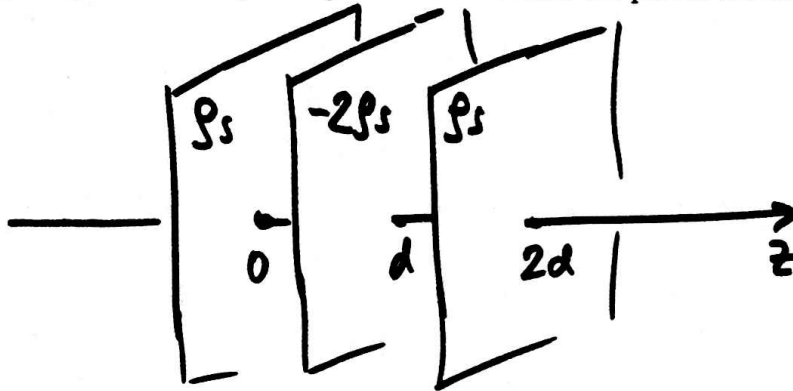
Problem 3: A sphere of radius a has uniform volume charge density ρ_v . There is no charge outside of the sphere. Assume the permittivity of the sphere and the surrounding medium is ϵ_0 .

a) Calculate $\mathbf{E}$ everywhere in space.

b) Calculate the electric potential V , assuming it is zero at an infinitely large distance from the sphere.

Problem 4:

Three parallel infinite planes have surface charge densities ρ_s , $-2\rho_s$, and ρ_s ($\rho_s > 0$). The spacing between adjacent planes is d . Assume the planes are in vacuum (permittivity ϵ_0).

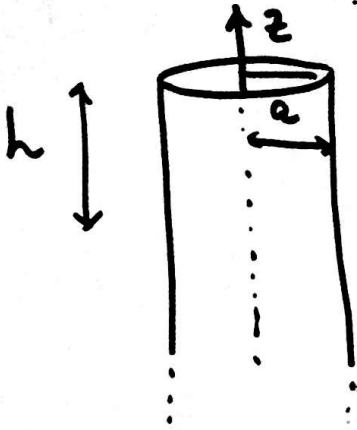


a) Calculate and graph the electric field $\mathbf{E}$ everywhere in space. Place the coordinate origin at the position of the leftmost plane, as in the figure above.

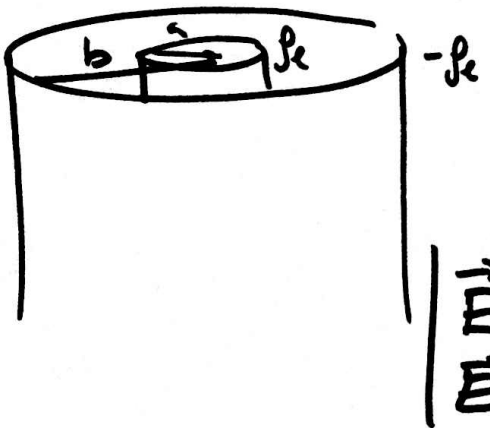
b) Calculate and graph the electric potential V everywhere in space. As the reference point, use the leftmost plane, i.e., assume $V(0)=0$.

Problem 5:

a) A cylindrical shell of line charge density ρ_l has radius a . Use Gauss's law to calculate the electric field $\mathbf{E}$ everywhere in space. Assume the surrounding medium is air (permittivity ϵ_0).



b) Another cylindrical shell, with the line charge density of $-\rho_l$ and radius $b > a$ is placed coaxially around the shell from part a). Based on your result from a) and using the principle of superposition, calculate and graph the total electric field everywhere in space.



c) Calculate the potential difference between the inner and outer shells, i.e., $V(a) - V(b)$.